

hw2

April 23, 2026

```
[29]: import numpy as np
import matplotlib.pyplot as plt
```

1 7

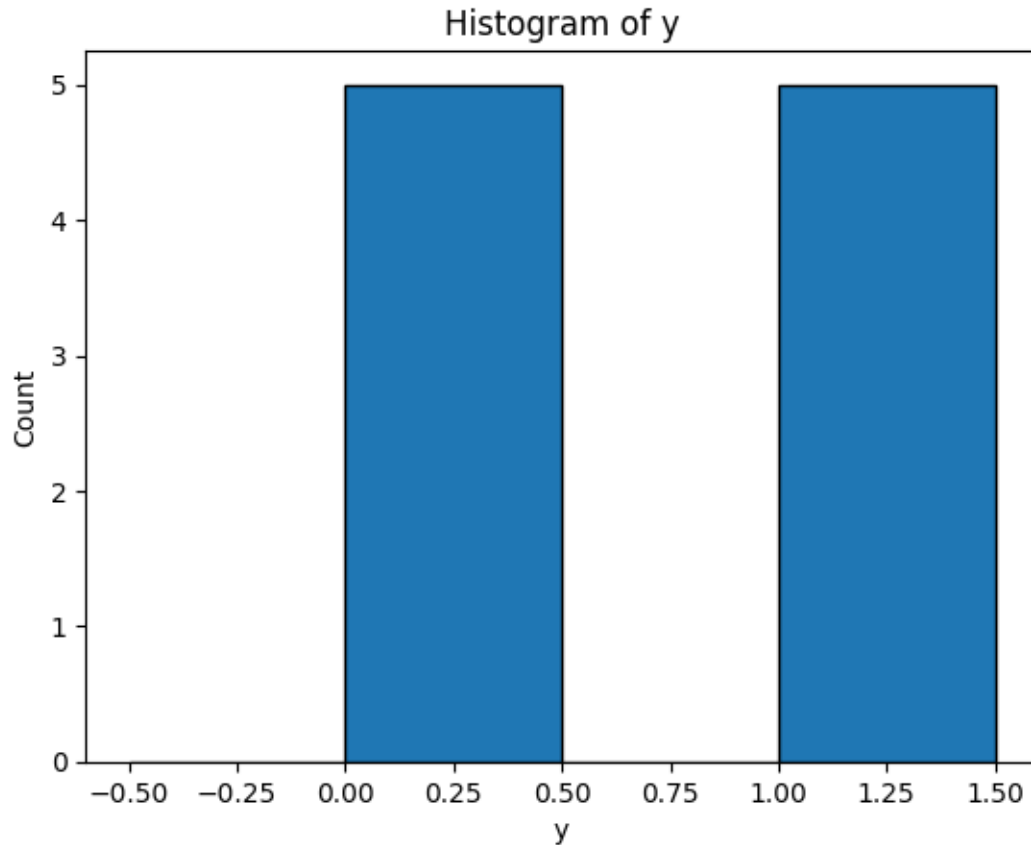
1.1 (a)

```
[30]: y = np.array([1, 0, 0, 0, 1, 0, 1, 1, 0, 1])
X = y.mean()
print(f"The movie x[11] is recommended to you, based on the fact that you liked_
↳{X*100}% of the comedies you watched previously")
```

The movie x[11] is recommended to you, based on the fact that you liked 50.0% of the comedies you watched previously

1.2 (b)

```
[31]: plt.hist(y,bins=4, range = (-0.5,1.5),edgecolor='black')
plt.xlabel('y')
plt.ylabel('Count')
plt.title('Histogram of y')
plt.show()
```



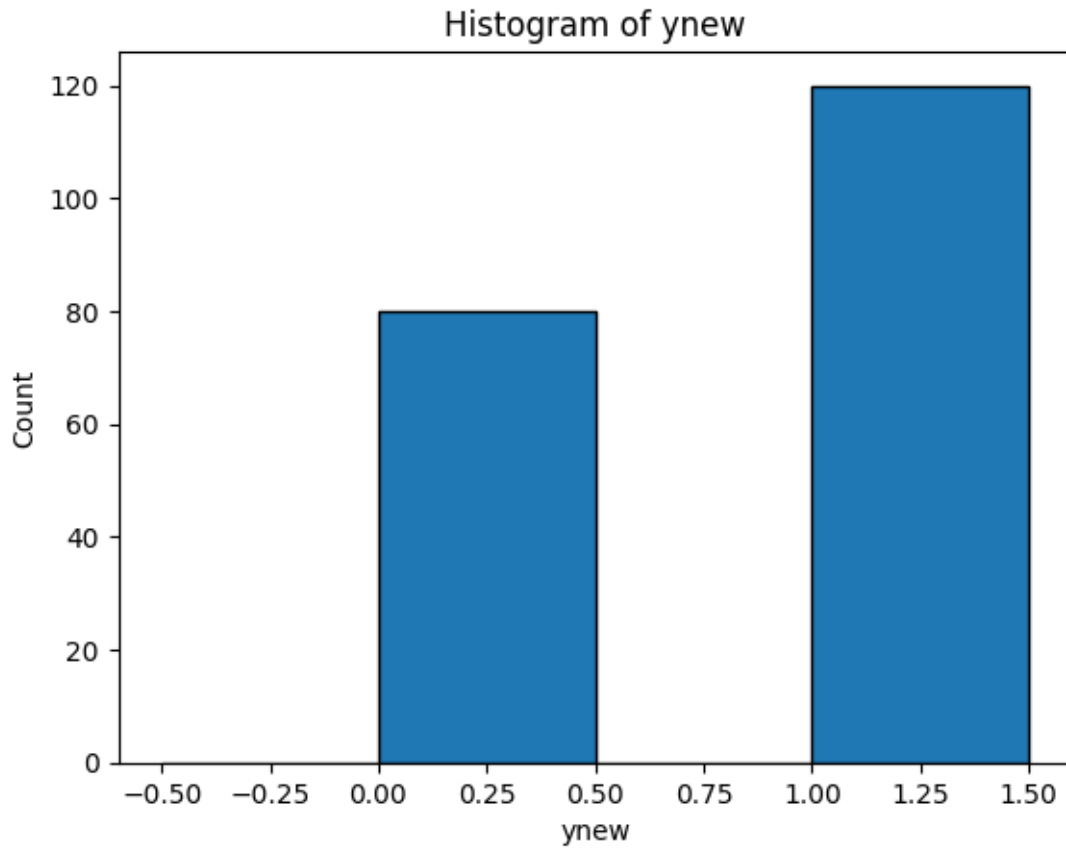
1.3 (c)

```
[32]: ynew = np.array([0, 1, 0, 1, 1, 0, 0, 1, 1, 0, 0, 1, 0, 1, 1, 1, 0, 0, 0, 0, 1,
↪0,
0, 1, 1, 0, 1, 1, 1, 1, 0, 1, 0, 0, 1, 0, 1, 0, 1, 0, 1, 1, 1, 0,
0, 1, 0, 1, 1, 1, 1, 1, 1, 1, 1, 0, 0, 0, 1, 0, 1, 1, 1, 1, 0, 1, 1, 1,
0, 1, 0, 0, 1, 1, 0, 1, 1, 0, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 0, 1, 1,
0, 0, 0, 1, 0, 1, 1, 0, 1, 1, 0, 1, 1, 1, 0, 1, 1, 0, 1, 1, 1, 0, 1,
1, 1, 0, 1, 0, 1, 1, 1, 0, 1, 0, 0, 1, 1, 0, 1, 0, 1, 1, 0, 1, 0, 1, 1,
1, 1, 0, 0, 1, 1, 1, 1, 0, 1, 1, 0, 0, 0, 1, 1, 0, 0, 1, 0, 1, 0, 1, 1,
1, 1, 1, 1, 1, 1, 1, 0, 0, 0, 1, 1, 1, 1, 0, 0, 1, 0, 1, 1, 1, 1, 0, 1])

ynew = np.concatenate((y, ynew))
Xnew = ynew.mean()
print(f"The movie x[11] is recommended to you, based on the fact that you liked_
↪{Xnew*100}% of the comedies you watched previously")
plt.hist(ynew, bins=4, range = (-0.5, 1.5), edgecolor='black')
plt.xlabel('ynew')
plt.ylabel('Count')
```

```
plt.title('Histogram of ynew')
plt.show()
```

The movie `x[11]` is recommended to you, based on the fact that you liked 60.0% of the comedies you watched previously



1.4 (d)

$$P(Y = 0) \approx 0.4, \quad P(Y = 1) \approx 0.6$$

1.5 (e)

```
[33]: z = np.array([0.77996466, 0.84599192, 0.60156913, 0.57652313, 1.0, 0.43186927,
↪ 0.67327696, 0.68174484, 0.8262356 , 1.0 ])
mean_z = np.mean(z)
print("Mean rating =", mean_z)
## with criteria as before. If the score is more than 0.5 then recommend.
if mean_z > 0.5:
    print("Recommendation decision: Recommend x[11]")
else:
```

```
print("Recommendation decision: Do not recommend x[11]")
```

Mean rating = 0.741717551

Recommendation decision: Recommend x[11]

2 8

2.1 (a)

```
[34]: import string
alphabet = string.ascii_letters
```

2.2 (b)

```
[35]: def counter(input_string:str)-> dict:
    letter_dict = {}
    for letter in input_string:
        if letter in alphabet:
            if letter in letter_dict:
                letter_dict[letter] += 1
            else:
                letter_dict[letter] = 1
    return letter_dict
```

2.3 (c)

```
[36]: c_dict=counter("Jim quickly realized that the beautiful gowns are expensive")
print(f'There are {c_dict["c"]} c and {c_dict["t"]} t in the sentence.')
```

There are 1 c and 4 t in the sentence.

2.4 (d)

```
[37]: address = """Four score and seven years ago our fathers brought forth on this_
    ↪continent,
a new nation, conceived in Liberty, and dedicated to the proposition that all_
    ↪men
are created equal. Now we are engaged in a great civil war, testing whether that
nation, or any nation so conceived and so dedicated, can long endure. We are met
on a great battle-field of that war. We have come to dedicate a portion of that_
    ↪field,
as a final resting place for those who here gave their lives that that nation_
    ↪might
live. It is altogether fitting and proper that we should do this. But, in a_
    ↪larger
sense, we can not dedicate -- we can not consecrate -- we can not hallow --_
    ↪this ground.
```

```

The brave men, living and dead, who struggled here, have consecrated it, far_
    ↪above
our poor power to add or detract. The world will little note, nor long remember_
    ↪what
we say here, but it can never forget what they did here. It is for us the_
    ↪living,
rather, to be dedicated here to the unfinished work which they who fought here_
    ↪have
thus far so nobly advanced. It is rather for us to be here dedicated to the_
    ↪great
task remaining before us -- that from these honored dead we take increased_
    ↪devotion
to that cause for which they gave the last full measure of devotion -- that we_
    ↪here
highly resolve that these dead shall not have died in vain -- that this nation,_
    ↪under
God, shall have a new birth of freedom -- and that government of the people, by_
    ↪the
people, for the people, shall not perish from the earth."""
d_dict = counter(address)
print(f'There are {d_dict["h"]} h in the paragraph.')

```

There are 80 h in the paragraph.

2.5 (e)

```

[38]: d_dict = counter(address)

most_freq = max(d_dict, key=d_dict.get)

print(f"The most frequent letter in the Gettysburg Address is {most_freq}, and_
    ↪it appears {d_dict[most_freq]} times.")

```

The most frequent letter in the Gettysburg Address is e, and it appears 165 times.